MATH 101

Chapter 3 Homework

Functions, composition, and inverses

Sections 3.1-3.7 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Give exact answers. Use a basic four-function calculator only if needed; show algebraic reasoning.

Function language and domains

1. Which relation defines y as a function of x? Which is also one-to-one? Explain.

$A = \{(-1, 2), (0, 2), (1, 4)\}$

$B = \{(-1, 2), (-1, 3), (1, 4)\}$

2. Find f(-2), f(a + 1), and the inputs for which f(x) = 5.

$f(x) = x^2 - 4$

3. Find the real domain of each function in interval notation.

a) $f(x) = \frac{x+2}{x^2-9}$

b) $g(x) = \sqrt{5 - 2x}$

4. For h(x) = -x + 1 when x < 0 and h(x) = x squared when x ≥ 0, find h(-2), h(0), and h(3). Sketch the two pieces, showing endpoint inclusion.

Rates and function structure

5. Find the average rate of change on [-1, 2]. Is the function constant on this interval? Explain.

$f(x) = (x - 1)^2$

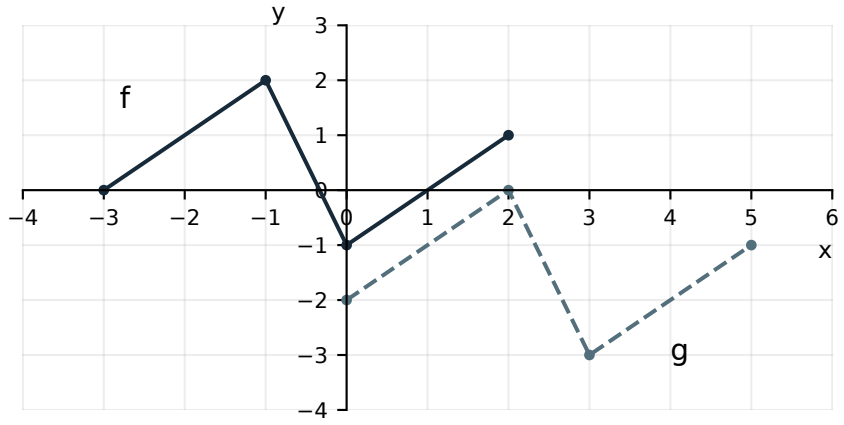
6. Find both compositions and their domains. Does order matter?

$f(x) = x^2, \qquad g(x) = x - 3$

7. Give functions f and g with h = f composed with g, with neither chosen function equal to the identity.

$h(x) = \sqrt{3x + 4}$

8. The solid graph is f and the dashed graph is g. Write g in terms of f. Read the domain and range of each graph.



Graph features and inverses

9. Describe the transformations from y = |x|. Sketch the graph and give its vertex, range, and increasing/decreasing intervals.

$$y = -2|x + 1| + 4$$

10. Find the inverse and verify both compositions.

$$f(x) = 3x - 5$$

11. Find the inverse of the restricted function. State the domain and range of the function and its inverse; explain the chosen square-root sign.

$$f(x) = (x - 2)^2, \qquad x \geq 2$$